

CSC15S Assignment 10 – Security Model and NFS

Total: 100pts (due date see Blackboard)

What to turn in: zip and upload all your related files to **Assignment 10** in Blackboard. Your zip should include the following files:

- (1) CSC15Slab10.docx (with all questions completed)
- (2) log_Mission8.txt

Part 1. Access Control Model and Session Management.

[20 pts. Estimated time to complete: 10 minutes.]

1. Give one scenario that the MAC model is better than the DAC model. [10 pts]

In a military setting, MAC is better for only users with appropriate clearance can access classified data, while DAC allows to grant access freely, increasing risk.

2. Session management can be implemented without the use of HTTP cookies. Describe how and compare it with the cookie approach (pros and cons)? [10 pts]

URL based session IDs can be used, where the session ID is included in the URL.

Pros: No reliance on cookies & simpler setup

Cons: Vulnerable to interception, less secure, prone to session hijacking

Cookie based sessions:

Pros: more secure (with httponly, secure flags).

Cons: depends on browser settings; cookies may be blocked.

Part 2. Dr. Evil's Training Room. Mission 8 – Godfather Part III

[80 pts. Estimated time to complete: 1.5 hrs]

Michael Corleone's new mission to modernize its empire is to allow NFS (Network File System) for everyone in his family. Help him to accomplish the goal!

Please note that you will apply changes to both the **Main** (169.254.251.2) and **Vegas** image (169.254.251.1). Misconfiguration can easily hang the entire OS. It's a good idea to take snapshots before addressing each of the five challenges (each might take about 200MB of disk space).

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Mission 8: Godfather III (80 pts)
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Name: Sean Fu
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Grade: 80.0 pts
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Time Spent: 404 seconds
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QID  Title                Pts    Time    QID  Title                Pts    Time
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1    Single Sign On        15.0/15 63     2    NFS Server           15.0/15 2
3    Home                   15.0/15 117    4    NFS Client           20.0/20 221
5    The Arts of Google     15.0/15 1
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Enter question ID (e.g., 1) to redo a question or hit Q to quit
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The following are some additional hints for each of the challenges.

Challenge	Hint
1. SS0	It's to replicate the last question of As9 . To verify, simply ' su mary ' in the main image and see if works (i.e., authenticating by LDAP instead of checking <code>/etc/passwd</code>)
2. NFS Server	Just follow the tutorial given. Simply some VIM editing and <code>apt-get</code> commands.
3. Home	Create the folders as specified. The trick is how to set the permission bits. Also, don't forget to chown!
4. NFS Client	To refresh after <code>/etc/fstab</code> is changed, do ' sudo mount -a '
5. Arts of Google	We'll leave the details to you. The trick is to find out how to properly use ldapmodifyuser to modify the homeDirectory attribute.